AWS SageMaker Clarify: Detecting data bias

Amazon SageMaker has a few new features to handle data preparation and analysis in a more efficient and intuitive manner. SageMaker Clarify is an inbuilt utility, which can be invoked from Amazon SageMaker Studio using the *Data flow -> Add Analysis* wizard. Amazon SageMaker Clarify can use binary and textual analysis to identify if the data has bias (right variances and grouping of data elements).

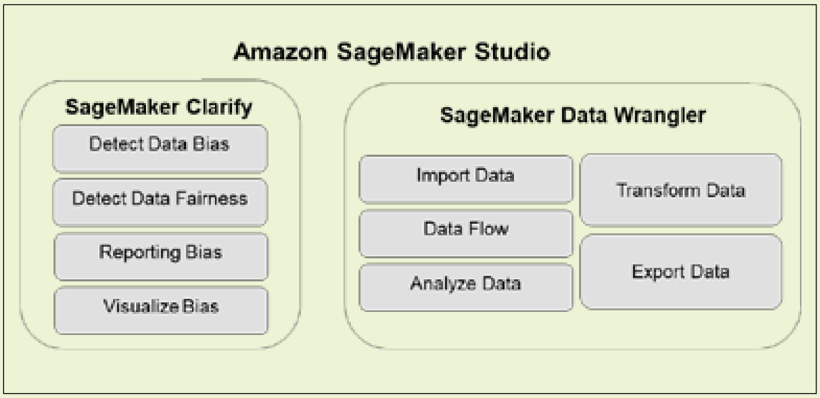

Figure 2: Component view of Amazon SageMaker Studio

Data fairness can be measured using fairness criteria, which are configurable and involve a rule-based setup in Studio notebook. Positive label value and negative label value are used for reporting bias and giving visual reports on data bias density.

A new feature introduced recently called ‘Data Wrangler’ enables end-to-end visual features for data import, data flow and transformation, data analysis and data export in ML enabled datasets to enable accurate and faster data analysis and processing. Data Wrangler has more than 300 data transformation templates in Pyspark 3 and Pandas or Pyspark (SQL) for data transformation in data science processing.

AI by demonstrating its ability to treat everyone fairly.

Challenges: Achieving fairness in AI is complex. Biases can be embedded in the training data, the algorithms themselves, and even in the way AI systems are used.

Accountability in AI

Definition: Refers to the ability to identify and hold someone or something responsible for the decisions and actions of an AI system. This includes understanding how the system arrived at a specific decision and who is accountable for any negative consequences.

Importance: Crucial for building trust and ensuring that AI systems are used responsibly. It allows for corrective actions to be taken in case of errors or biases.

Challenges: Establishing accountability for AI systems can be challenging due to their complex nature and the often opaque decision-making processes.

The intersection of fairness and accountability

Fairness and accountability are interconnected. Achieving fairness in AI requires holding someone accountable for potential biases in the system; conversely, lacking accountability makes it difficult to ensure fairness.

Here are some strategies to help achieve fairness in an AIML model.

Transparency: Implementing techniques like explainable AI (XAI) to understand how AI systems reach decisions, fostering transparency and identifying potential biases.

Auditing and monitoring: Regularly auditing AI systems for bias and unintended consequences.

Clear governance frameworks: Establishing clear guidelines and regulations for responsible AI development and deployment, including well-defined roles and responsibilities.

By addressing both fairness and accountability, we can strive towards building AI systems that are not only powerful and efficient but also equitable and trustworthy. This requires ongoing collaboration between developers, policymakers, and the public to develop and implement responsible AI practices.

Industry standards and guidelines for ethical AI

Fortunately, several industry standards and guidelines have emerged to guide developers and users in building and utilising AI responsibly.

- **The European Union (EU):** The EU is playing a prominent role in shaping ethical AI practices. Its ‘Ethics Guidelines for Trustworthy AI’ emphasise seven key requirements.

1. ***Human agency and oversight:*** AI systems should be under human control and supervision.
2. ***Technical robustness and safety:*** AI systems should be robust, reliable, and secure to prevent harm.
3. ***Privacy and data governance:*** Personal data used in AI development and deployment needs to be handled ethically and responsibly.
4. ***Transparency:*** AI systems and their decision-making processes should be transparent and understandable.